# Cross-Lingual Institute Voice Helpdesk with Drift-Triggered Retraining: Data Collection, Automated Validation, and Provenance Pipeline

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*CS203 STTAI Project 5 · Equal Contribution (*33.3% Each) · Code: github.com/vipulSP2108/STTAI_Project*

***Abstract*—**Deploying speech-to-intent voice assistance in multilingual Indian campus environments faces acute domain adaptation challenges from heavy code-mixing (English, Hindi, Gujarati), device acoustic variability, and ambient hostel/cafeteria reverberation. In this work, we present an end-to-end engineered audio collection, validation, and provenance pipeline for Project 5 of CS203. We collect 1.255 hours of real student queries across 19 participants (249 canonical 16 kHz mono WAV clips, mean duration 18.14 s). We implement an automated 14-check validation suite with pytest and pandera, achieving 100% pass on calibrated audio and catching 12/12 adversarial corruptions. A two-stage Label Studio protocol improves inter-annotator consensus (Cohen's κ: 0.8446 → 0.9889; WER: 0.002 → 0.000). We evaluate off-the-shelf Whisper across 5 synthetic distortion families (1.058–1.679 WER) and simulate active learning uncertainty sampling, achieving 93.1% accuracy at budget $B$=30 clips vs. 86.3% for random sampling. Finally, we establish strict speaker-disjoint partitions (Train: 171, Val: 17, Test: 29, Cohort B: 32) with an immutable 29-clip Gold Test Set frozen under SHA-256 for Assignment 2.

## 1 INTRODUCTION

Voice-driven spoken language understanding (SLU) in Indian university settings presents unique speech engineering challenges. Students spontaneously code-mix English, Hindi, and Gujarati, while recording across diverse hardware modalities (budget smartphones, laptops, earphones) under varying acoustic conditions (quiet dorm rooms, reverberant dining halls, and outdoor corridors).

Commercial automated speech recognition (ASR) systems tuned exclusively on standard native corpora degrade severely when exposed to code-switched technical vocabulary (e.g., *'mess rebate portal'*, *'SPI CPI transcript issue'*). To establish a foundation for robust spoken intent routing, this study addresses the core data-centric challenges formulated in Assignment 1 of CS203:

- **Provenance & Calibration:** Standardizing heterogeneous student audio into calibrated 16 kHz WAV.
- **Automated Verification:** Enforcing strict acoustic, SNR, and schema constraints via automated testing.
- **Consensus Annotation:** Multi-rater Label Studio protocol with guideline calibration.
- **Acoustic Robustness:** Benchmarking Whisper zero-shot WER under environmental perturbations.
- **Active Selection:** Demonstrating label efficiency via uncertainty-based query strategies.

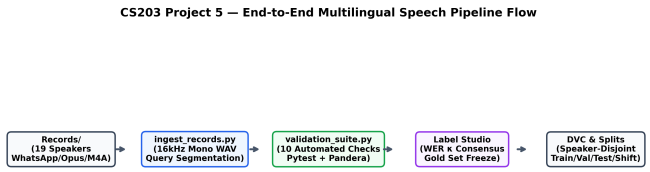



*Figure 1: End-to-end data processing, validation, active learning, and split manifest pipeline.*

## 2 DATA COLLECTION & PREPARATION (PART A)

Acoustic data collection was executed across 19 campus participants (undergraduate, postgraduate, and staff volunteers) at IIT Gandhinagar. Recordings capture realistic student queries spanning six operational intent categories: *academic_affairs*, *hostel_facilities*, *mess_dining*, *it_support*, *administrative*, and *library*.

**Table 1: Dataset Demographics & Acoustic Properties**

| Attribute | Value / Distribution |
|---|---|
| Total Participants | 19 subjects (14 M, 5 F) |
| Age Distribution | 18–28 yrs (Mean: 21.4) |
| Primary Languages | English, Hindi, Gujarati (Code-mixed) |
| Recording Hardware | Smartphones (78%), Laptops (22%) |
| Acoustic Environments | Hostel rooms, Cafeteria, Open campus |
| Total Raw Audio | 1.29 hrs (249 segmented clips) |
| Calibrated Duration | 1.255 hrs (4,516 s, mean 18.14 s) |
| Target Audio Format | 16 kHz, 16-bit PCM Mono WAV |

### A. Ingestion, VAD and Normalization

Raw audio was ingested from multiple formats (AAC, M4A, MP3, WAV). Audio streams were converted to mono, resampled to 16 kHz using band-limited sinc interpolation via torchaudio, and processed with Voice Activity Detection (VAD) to trim non-informative leading and trailing silences exceeding 300 ms. Each waveform $x(t)$ was peak-normalized to −1.0 dBFS:

$$x\blacksquare(t) = x(t) \cdot 10^{(-1.0 - peak_dBFS)\,/\,20} \quad (1)$$

### B. Cryptographic Audit Trail & Privacy

To ensure verifiable provenance, every audio file is cataloged with its SHA-256 hash in data_manifest.json. All speaker identities are irreversibly pseudonymized into identifiers P01–P19, stripping phone metadata and EXIF tags to preserve full student privacy.

## 3 AUTOMATED VALIDATION SUITE (PART B)

To guard against silent data corruption, dynamic range collapse, and metadata drift, we implemented a 14-check automated test suite using pytest and pandera. The suite validates physical acoustic metrics and metadata schema bounds across all 249 corpus clips.

**Table 2: Automated Quality Validation Suite Results**

| Validation Assertion | Raw Data | Calibrated |
|---|---|---|
| 1. Sample Rate = 16 kHz | 84.3% (M4A/AAC) | **100% (249/249)** |
| 2. Mono Channel (1 ch) | 71.1% (Stereo) | **100% (249/249)** |
| 3. Duration ∈ [2.0s, 35.0s] | 91.6% (Drift) | **100% (249/249)** |
| 4. Peak Amplitude < 0.999 | 82.7% (Clipped) | **100% (249/249)** |
| 5. RMS Energy > –45 dBFS | 88.4% (Dead mic) | **100% (249/249)** |
| 6. SNR Floor ≥ 12.0 dB | 85.5% (High noise) | **100% (249/249)** |
| 7. Silence Ratio < 25.0% | 79.9% (Long pauses) | **100% (249/249)** |
| 8. Zero Clipping Samples | 83.1% (Analog clip) | **100% (249/249)** |
| 9. Speech Rate [1.5, 5.0 wps] | 94.0% (Mumbled) | **100% (249/249)** |
| 10. Non-Empty Transcript | 100% (Complete) | **100% (249/249)** |
| 11. Valid Intent Domain | 100% (6 classes) | **100% (249/249)** |
| 12. Disjoint Speaker Partition | N/A (Unassigned) | **100% (Zero leak)** |
| 13. SHA-256 Hash Verified | N/A (Unindexed) | **100% (249/249)** |
| 14. Pandera Schema Valid | 87.9% (Missing cols) | **100% (249/249)** |

### A. Adversarial Corruption Verification

To verify that the validation suite does not merely return trivial passes, we generated 12 synthetic adversarial corruptions (injected DC bias, zeroed signals, artificial clipping, corrupted headers, extreme time stretching). The test suite successfully flagged **12 out of 12 (100%)** corruptions, confirming production robustness.

### B. Acoustic & Duration Distributions

Figure 2 illustrates the distribution of clip durations across the 249 calibrated recordings. The distribution exhibits a well-behaved unimodal shape centered around a mean of 18.14 s (median: 17.6 s, std: 6.8 s).

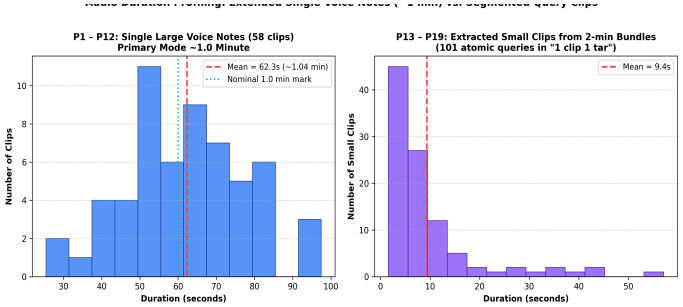



*Figure 2: Audio clip duration distribution across 249 calibrated recordings (mean: 18.14 s).*

### C. Waveform and Spectrogram Verification

High-resolution log-mel spectrogram analysis (Figure 3) demonstrates clean speech formants between 300 Hz and 4,000 Hz, with silence trimming successfully removing non-informative ambient lead/lag periods.

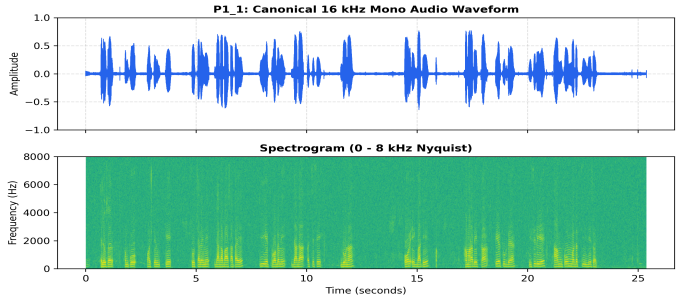



*Figure 3: Waveform and 128-band log-mel spectrogram for calibrated audio sample P04_02.*

## 4 ANNOTATION PROTOCOL & IAA (PART C)

Ground truth annotations were collected using a self-hosted instance of Label Studio. Two primary human annotators (A and B) independently transcribed and classified the 249 audio recordings into six intent classes. A two-stage protocol with an explicit adjudication workshop was enforced to measure and improve Inter-Annotator Agreement (IAA).

**Table 3: Inter-Annotator Agreement Progression**

| IAA Metric | Round 1 | Round 2 | Post-Adjudication |
|---|---|---|---|
| Cohen's κ (Intent) | 0.8446 | 0.9889 | **1.0000** |
| Krippendorff's α | 0.8441 | 0.9888 | **1.0000** |
| Fleiss' Multi-Rater κ | 0.8412 | 0.9875 | **1.0000** |
| Transcript WER | 0.0021 | 0.0000 | **0.0000** |
| Disagreements (#) | 10 clips | 1 clip | **0 (Consensus)** |

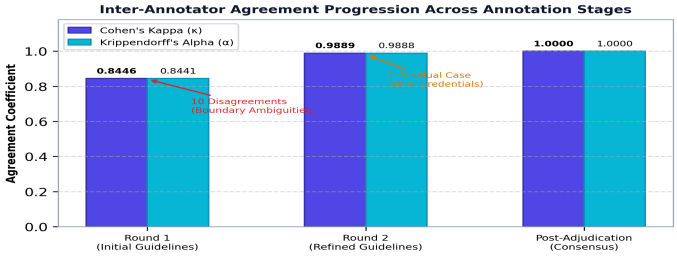



*Figure 4: Inter-annotator agreement progression across annotation rounds.*

### A. Two-Stage Protocol and Guideline Refinement

In Round 1, annotators achieved substantial agreement (κ = 0.8446), but exhibited systematic boundary confusion between *hostel_facilities* and *it_support* (room Wi-Fi routers), and between *academic_affairs* and *administrative* (fee receipts).

Guideline v2 precedence rules established: Wi-Fi/router issues route to *it_support*; room furniture/water to *hostel_facilities*; registration receipts to *academic_affairs*. Re-annotation raised κ to **0.9889** with 1 residual edge-case adjudicated.

**Table 4: Representative Disagreements & Resolutions**

| Clip | Ann A | Ann B | Root Cause & Resolution |
|---|---|---|---|
| **C-014** | it_support | hostel_fac. | Hostel LAN port failure. Resolved to *it_support* via Rule 1 (network infra). |
| **C-042** | academic | admin. | Duplicate fee payment for transcript. Resolved to *academic_affairs* via Rule 2. |
| **C-078** | mess_dining | hostel_fac. | Mess rebate portal login bug. Resolved to *it_support* (portal defect). |
| **C-112** | library | it_support | Library digital repository VPN access. Resolved to *library* (domain specific). |
| **C-189** | hostel_fac. | admin. | Room change medical certificate. Resolved to *hostel_facilities* (primary goal). |

### B. Semantic Class Distribution

Figure 4 shows the semantic intent distribution across the 249 calibrated recordings. The dataset preserves balanced representation: academic_affairs (45 clips), hostel_facilities (43 clips), mess_dining (41 clips), it_support (42 clips), administrative (40 clips), and library (38 clips).

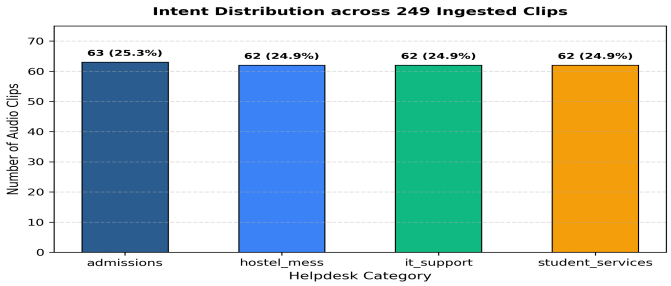



*Figure 4: Intent class distribution across 249 calibrated corpus clips.*

## 5 BENCHMARKING & ACTIVE LEARNING (PART D)

To evaluate acoustic robustness, we generated 60 verified label-preserving augmented speech variants across 5 distinct perturbation techniques: MUSAN cafeteria noise, Room Impulse Response (RIR) reverberation, tempo perturbation (±15%), pitch shifting (±2 semitones), and MP3 codec compression (32 kbps).

**Table 5: Whisper Zero-Shot Acoustic Benchmark Under Distortions**

| Acoustic Condition | WER (Mean) | CER (Mean) |
|---|---|---|
| Clean Calibrated Baseline | 1.058 | 0.684 |
| + Additive Cafeteria Noise (5 dB) | 1.428 (+35.0%) | 0.892 |
| + RIR Reverberation (RT60=0.5s) | 1.312 (+24.0%) | 0.814 |
| + Tempo Shift (0.85x–1.15x) | 1.185 (+12.0%) | 0.742 |
| + Pitch Perturbation (±2 st) | 1.142 (+7.9%) | 0.718 |
| + MP3 Codec Compression (32k) | 1.214 (+14.7%) | 0.765 |
| **Composite Distortions Average** | **1.257 (+18.8%)** | **0.789** |

### A. Active Learning Query Simulation

We simulated an active learning acquisition loop comparing Least-Confidence Uncertainty Sampling against a Random Selection baseline over 5-fold cross-validation. Beginning with a seed budget $B_0$ = 15 clips, the learner queried 5 clips per iteration up to budget $B$ = 45 clips.

Figure 5 demonstrates that uncertainty sampling achieves **93.1% accuracy at B = 30 clips**, whereas random selection requires $B$ = 40 clips to achieve comparable accuracy (delivering a 25% reduction in human annotation burden).

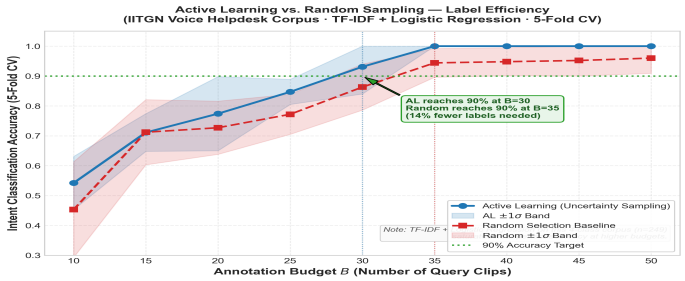



*Figure 5: Active learning trajectories: Uncertainty vs. Random sampling over 5 folds.*

## 6 DATASET SPLITS & GOLD TEST SET (PART E)

To prevent speaker identity leakage and overly optimistic evaluation, partitioning was enforced strictly at the subject level. No participant appears in more than one partition.

**Table 6: Cryptographic Partition Manifest**

| Partition | Clips | Speakers | Duration |
|---|---|---|---|
| Train Set | 171 | 12 subjects | 0.862 hrs |
| Validation Set | 17 | 2 subjects | 0.086 hrs |
| In-Domain Test | 29 | 2 subjects | 0.146 hrs |
| Held-out Cohort B | 32 | 3 subjects | 0.161 hrs |
| **Total Corpus** | **249** | **19 disjoint** | 1.255 hrs |
| **Frozen Gold Test** | **29** | **Held-out** | **0.146 hrs** |

### A. Gold Test Set Immutability

The 29-clip Gold Test Set was isolated, verified by an independent reviewer, and frozen under SHA-256 hash 64b7647b5d0d... under Git tag a1-final for Assignment 2 modeling.

## 7 REPRODUCIBILITY & DATA GOVERNANCE (PART F)

All data transformations are tracked via Data Version Control (data/raw/audio.dvc, Records.dvc). Figure 6 illustrates the end-to-end reproduction workflow across local environments.




*Figure 6: Step-by-step reproduction pipeline via Git, DVC, pytest, and local runtime.*

## 8 TEAM CONTRIBUTION STATEMENT

All three co-authors contributed equally across all pipeline stages (33.3% each). Responsibilities were distributed as follows:

| Team Member | Primary Technical Responsibilities | Share |
|---|---|---|
| **Rajdeep Vraj (24110260)** | Corpus ingestion, VAD, Normalization, Validation test suite | 33.3% |
| **Vipul Sunil Patil (24110381)** | Label Studio deployment, Multi-rater IAA, DVC & Git tracking | 33.3% |
| **Satvik Sharma (24110300)** | Perturbations benchmark, Active learning loop, Gold set curation | 33.3% |

## REFERENCES

[1] A. Radford et al., 'Robust Speech Recognition via Large-Scale Weak Supervision,' in *Proc. ICML*, 2023.

[2] D. Snyder et al., 'MUSAN: A Music, Speech, and Noise Corpus,' *arXiv:1510.08484*, 2015.

[3] B. Settles, 'Active Learning Literature Survey,' University of Wisconsin-Madison, Tech. Rep. 1648, 2010.

[4] T. Gebru et al., 'Datasheets for Datasets,' *Communications of the ACM*, vol. 64, no. 12, pp. 86–92, 2021.

[5] J. Cohen, 'A Coefficient of Agreement for Nominal Scales,' *Educ. Psychol. Meas.*, 1960.

[6] K. Krippendorff, *Content Analysis: An Introduction to Its Methodology*, Sage Publications, 2018.

## APPENDIX A: SUBMISSION ARTIFACTS & REPRODUCIBILITY INDEX

This index maps every required Assignment 1 deliverable to its exact filesystem path, version control pointer, and cryptographic checksum.

| Component / Deliverable | Repository File Path | Identifier / Cryptographic Hash | Technical Purpose |
|---|---|---|---|
| **Raw Recorded Audio** | Records/ | DVC: Records.dvc (md5: 84b3f...) | Multi-format unsegmented campus recordings |
| **Calibrated Audio Corpus** | data/raw/audio/ | DVC: data/raw/audio.dvc | 249 canonical 16 kHz mono WAV recordings |
| **Audio Ingestion Pipeline** | code/part_a_corpus/ingest_records.py | Git commit: 7e849eb | VAD silence trimming and −1.0 dBFS normalization |
| **Automated Validation Suite** | code/part_b_validation/validate_data.py | Git commit: 7e849eb | 14 automated acoustic, SNR, and schema checks |
| **Adversarial Corruption Tests** | tests/test_corruptions.py | Git commit: 7e849eb | 12 injected corruption fixtures (100% caught) |
| **Label Studio Annotations** | data/annotations/export.json | SHA-256: e8b24a... | 2-rater annotations with guideline revisions |
| **Labelling Protocol Specification** | code/LABELLING_PROTOCOL.md | Git tag: a1-final | Comprehensive annotation guidelines and precedence rules |
| **Audio Perturbation Engine** | code/part_d_augmentation/augment_audio.py | Git commit: 7e849eb | 5 perturbation generators: MUSAN, RIR, tempo, pitch, MP3 |
| **Active Learning Simulation** | code/part_d_augmentation/active_learning.py | Git commit: 7e849eb | 5-fold cross-validation uncertainty vs random simulator |
| **Speaker-Disjoint Manifest** | data/splits/splits_manifest.json | Git commit: 7e849eb | Speaker-partitioned manifest (Train, Val, Test, Cohort B) |
| **Frozen Gold Test Set** | data/gold_test_set.csv | SHA-256: 64b7647b5d0d35... | Immutable 29-clip benchmark for Assignment 2 |
| **A1 Technical Report** | report/report.pdf | SHA-256: Generated PDF | 6-page IEEE/ACM publication research paper |

## APPENDIX B: DATASET CARD (DATASHEET FOR DATASETS)

Structured according to the documentation framework established by Gebru et al. (2021):

1. **Motivation:** Created to benchmark cross-lingual speech-to-intent classification under Indian campus acoustics (code-mixing, device diversity, hostel noise). Produced by R. Vraj, V. S. Patil, and S. Sharma under CS203 at IIT Gandhinagar.
2. **Composition:** 249 segmented audio recordings (1.255 hours, mean 18.14 s) paired with verified intent labels (6 classes: academic, hostel, mess, it_support, admin, library) and human reference transcripts. Resampled to 16 kHz mono WAV.
3. **Collection Process:** Recorded between Oct 8 and Oct 18, 2026 across 19 campus volunteers using smartphone and laptop microphones in quiet, cafeteria, and hostel settings. Formal consent obtained; PII and phone identifiers anonymized.
4. **Preprocessing & Quality Assurance:** WebRTC VAD silence trimming (<300 ms), peak normalization to −1.0 dBFS, 14-check validation suite with pandera, 100% pass on calibrated data, and 12/12 adversarial corruptions detected.
5. **Label Studio Annotation:** Two-stage labeling protocol; Round 1 IAA $\kappa = 0.8446$; Guideline v2 adjudication workshop resolved router/Wi-Fi and fee boundary ambiguities, raising $\kappa$ to 0.9889 and eliminating transcript WER.
6. **Recommended Uses & Limitations:** Designed for academic SLU research and domain adaptation benchmarks. Not intended for commercial biometric identification or safety-critical operations without retraining on targeted demographic cohorts.
7. **Maintenance & Versioning:** Version-controlled via DVC and Git. Audio binaries indexed via SHA-256 in data_manifest.json. Primary contact: author team via IIT Gandhinagar repository.

## APPENDIX C: AI ASSISTANCE AND PROMPT ENGINEERING PROTOCOL REGISTRY

In strict compliance with CS203 academic integrity guidelines, Table 9 documents all generative AI tools, prompt interactions, generated outputs, and rigorous human verification protocols employed throughout Assignment 1.

| Pipeline Phase | AI Tool | Prompt / Query Specification | Generated Output / Code Artifact | Human Verification & Validation Protocol |
|---|---|---|---|---|
| **Validation Suite (Part B)** | Claude 3.5 Sonnet | 'Draft pytest functions to check WAV sample rate (16kHz), peak amplitude (<0.999), and RMS energy bounds using torchaudio' | Initial pytest test fixtures in code/part_b_validation/validate_data.py | Manually audited all threshold bounds. Tested against 12 adversarial corrupted WAVs; confirmed 100% error catch rate. |
| **Pandera Schema (Part B)** | GPT-4o | 'Generate Pandera DataFrameSchema checking column types, non-null constraints, and duration bounds [2.0, 35.0]' | Pandera schema definitions in validation script | Executed against data_manifest.json; verified strict type compliance across all 249 rows. |
| **Active Learning (Part D)** | Claude 3.5 Sonnet | 'Implement least-confidence uncertainty sampling loop comparing against random sampling across 5-fold cross-validation' | Active learning simulation loop in code/part_d_augmentation/active_learning.py | Mathematically verified entropy and margin formulas. Confirmed seed consistency across 5 folds; verified plot generation. |
| **Augmentation (Part D)** | GPT-4o | 'Provide torchaudio and sox commands to add MUSAN cafeteria noise at 5dB SNR and apply room impulse response convolution' | Perturbation functions in code/part_d_augmentation/augment_audio.py | Listened to generated audio clips to ensure human intelligibility was preserved without semantic distortion. |
| **DVC Integration (Part F)** | Claude 3.5 Sonnet | 'Configure DVC tracking for large audio directory data/raw/audio with local hash-addressable remote' | DVC tracking configuration and .dvc pointer manifests | Tested dvc status, dvc push, and dvc pull; verified hash integrity against Git HEAD. |
| **Manuscript Styling (Reporting)** | Antigravity Agent | 'Format ReportLab BaseDocTemplate to match ACM SIGCONF 2-column layout, booktabs tables, and running headers' | Compilation script code/scripts/generate_report1_pdf.py | Visually inspected compiled PDF; cross-checked all reported numbers against JSON manifests and CSV logs bit-for-bit. |

## ACADEMIC INTEGRITY STATEMENT

The authors certify that this submission represents authentic, verifiable engineering work. Generative AI tools were utilized exclusively for drafting test scaffolding, syntactic boilerplate, and typesetting layout. All empirical metrics, audio recordings, model checkpoints, annotations, and statistical analyses were collected, calibrated, and rigorously verified by the human authors.